ECE 329 **Homework 13** **Due: Fri, Dec 26, 2025, 11:59PM**

4 problems totaling 30pts with 10pts extra Bonus Points. (Bonus problems are optional and the earned bonus credits will be used to "fill in" the "missing points" in the total homework score.)

1. **Bonus Problem: (10pts**) Consider a complex variable $z = r + jx$, where r and x are real variables. Also consider a transform of z given by the formula

$$\Gamma \equiv \frac{z-1}{z+1} = \frac{r+jx-1}{r+jx+1} = \frac{[(r-1)+jx][(r+1)-jx]}{(r+1)^2+x^2} = \frac{(r^2+x^2-1)+j2x}{(r+1)^2+x^2} \equiv \Gamma_r + j\Gamma_i.$$

From above, it follows that

$$\Gamma_r = \frac{(r^2+x^2-1)}{(r+1)^2+x^2} \text{ and } \Gamma_i = \frac{2x}{(r+1)^2+x^2}$$

are the real and imaginary parts of the transform Γ. Above, r and x denote the normalized line resistance and reactance, respectively, in a TL circuit, and Γ_r and Γ_i denote the real and imaginary parts of reflection coefficient Γ. Using the above expressions for Γ_r and Γ_i, it can be shown that the following relations are valid:

$$(\Gamma_r - \frac{r}{r+1})^2 + \Gamma_i^2 = (\frac{1}{r+1})^2 \quad \text{eqn for constant } r \text{ circles}$$

$$(\Gamma_r - 1)^2 + (\Gamma_i - \frac{1}{x})^2 = (\frac{1}{x})^2 \quad \text{eqn for constant } x \text{ circles}$$

a) **Smith Chart construction:** The equations above describe circles on the complex Γ-plane with r and x dependent centers and radii, respectively. Using a compass, draw constant r circles (on a plane with Γ_r and Γ_i axes) for the values of $r = 0$, 1, 2, ∞ — set the scales of your axes such that your circles comfortably fill a 8.5"X11" page, and mark each circle by the r-value. Also draw and mark portions of constant x circles overlapping the same region occupied by the constant r circles, for values of $x = \pm 0.5, \pm 1, \pm 2$. Finally, identify and mark the locus of $x = 0$ in the same region.

b) Download a Smith Chart (SC) from the course web site, and highlight and mark the same set of constant r and constant x circles drawn in part (a). Staple the SC with the rest of your HW.

Congratulations! You have developed a SC and have seen that the SC is the interior of unit circle on the complex plane gridded in a special way to recognize the mapping of all possible complex $z = r + jx$**,** $r \geq 0$**,** onto the unit circle and its interior. In the following, we will have several exercises demonstrating various uses of the SC.

2. (10pts) A TL with characteristic impedance $Z_o = 50\,\Omega$ and length $l = 0.2\lambda$ has a *short* termination at $d = 0$ and a $50\,\Omega$ resistor connected between the TL conductors (a "shunt" connection) at $d = 0.1\lambda$. Use a SC to determine the input impedance and admittance of the line at the generator end, i.e., $Z(l)$ and $Y(l)$. Hand in your marked SC.

   Hint: first move by 0.1λ from the load point (where you enter $z(0)$ or $y(0)$) toward generator on the SC, read off the corresponding $y(d)$, combine it in parallel with the shunt element, go back onto the SC with the normalized combined admittance, and move another 0.1λ toward the generator...

3. (10pts) We want to use a quarter-wave transformer to match a load Z_L to a T.L. with a characteristic impedance $Z_o = 50\,\Omega$. The quarter wave transformer is to be inserted at a distance $d_1$away from the load. Determine d_1 (in units of λ) and the characteristic impedance Z_{qo} of the quarter wave transformer if

   a) $Z_L = 100\,\Omega$,

   b) $Z_L = 100 + j100\,\Omega$.

4. (10pts) We want to use a single-stub tuner to match a load Z_L to a T.L. with a characteristic impedance $Z_o = 50\,\Omega$. A shorted-stub is to be connected at a distance $d_1$away from the load. Determine d_1 (in units of λ) and the length d_s of the shorted stub if $Z_L = 100 + j100\,\Omega$ and the characteristic impedance of the stub is $50\,\Omega$.